Course Code: EET205

Course Name: ANALOG ELECTRONICS

Max. Marks: 100 Duration: 3 Hours

PART A

Answer all questions. Each question carries 3 marks

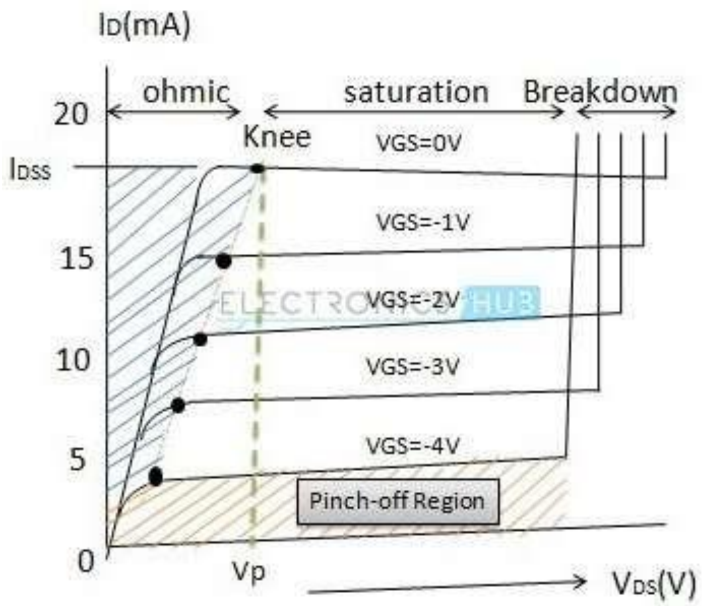
1 **(Few points):** (i) The operating point is so chosen such that it lies in the active region and proper selection of it helps in the faithful amplification of complete signal (without any loss). Impact of Q pt selection at centre (OR) near cut off point /saturation point to be mentioned. (3)

(iii)Affected by β, I_{CO}, V_{BE}

2 $I_B = 119\ \mu A$ $I_C = \beta I_B = 5.95\ mA$ (3)

$$V_{CE} = V_{CC} - I_C R_C = 4.3\ V$$

3 (3)



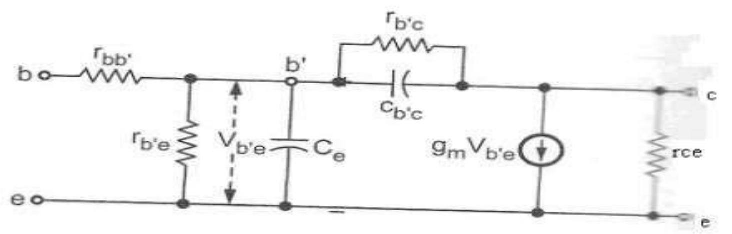
4  (3)

Fig. Hybrid - ∏ model for a transistor in CE configuration

5 The Class B Amplifier (Q-pt at cut off region) ➔ NO power is dissipated in the transistors when there is no signal present (unlike Class A) (3)

Drawback ➔ Cross over distortion & not faithful amplification (unlike Class A)

6 The Barkhausen criterion states that: (3)

• The loop gain is equal to unity in absolute magnitude, that is, $|\beta A| = 1$ and

• The phase shift around the loop is zero or an integer multiple of 2π (or 360^0)

7 CMRR - ratio of differential-mode voltage gain (Ad) and the common-mode voltage gain (3)

The slew rate - The maximum possible rate of change of output voltage of Op-Amp with respect to time

8.-- (3)

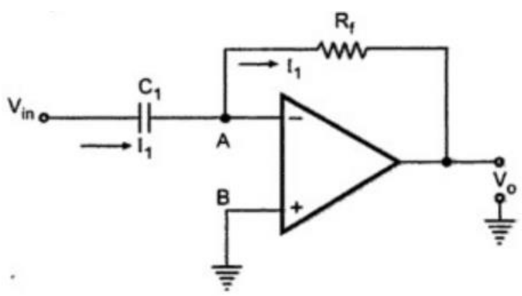
Infinite open-loop gain

- Infinitely high input impedance
- Zero input offset voltage
- Infinite output voltage gain
- Gain independent of frequency

(Infinite bandwidth with zero phase shift and infinite slew rate)

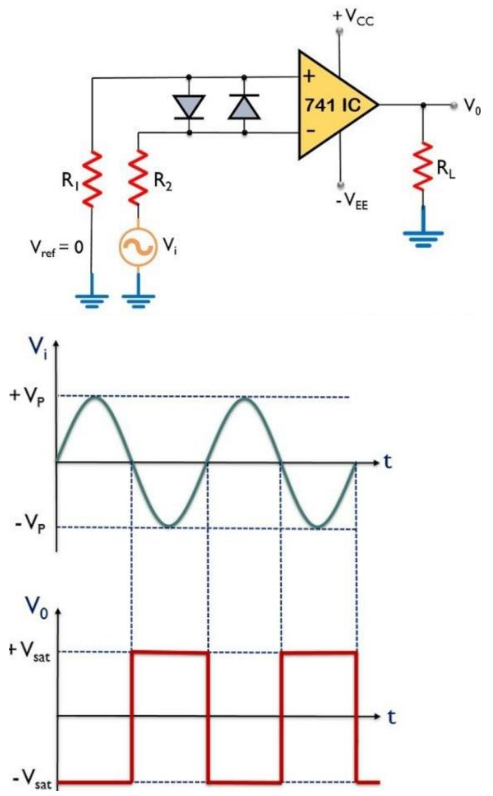
- Zero output impedance
- Zero noise
- Infinite common-mode rejection ratio (CMRR)

9



$$I_1 = C_1\left(\frac{d(V_{in}-V_A)}{dt}\right) = C_1\left(\frac{dV_{in}}{dt}\right)$$
$$I = \left(\frac{V_A - V_0}{R_f}\right) = -\left(\frac{V_0}{R_f}\right)$$
$$V_0 = -C_1 R_f\left(\frac{dV_{in}}{dt}\right)$$

10



As we have already discussed that it detects the point where the input signal crosses zero of the reference voltage level. For every crossing, the saturation level of the output signal changes from one to another.

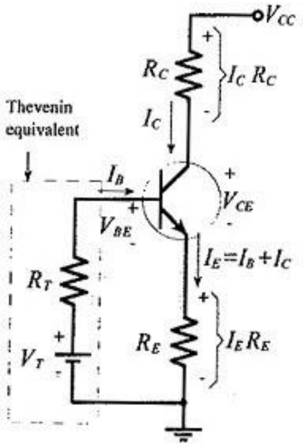
(3)

PART B

Answer any one full question from each module. Each question carries 14 marks

Module 1

11



Voltage divider bias with *Thevenin equivalent circuit* of the voltage divider

(a) Replace input with Thevenin eqivalent (9)

$$V_{\mathrm{T}} = V_{\mathrm{CC}} \frac{R_2}{R_1 + R_2} = 18X \frac{22k}{(82k + 22k)}$$

$$= 3.81\, V$$

$$R_{\mathrm{T}} = \frac{R_1 R_2}{R_1 + R_2} = \frac{82k\ X\ 22k}{(82k + 22k)} = 17.35\ k\Omega$$

$$V_{\mathrm{T}} = I_{\mathrm{B}} R_{\mathrm{T}} + V_{\mathrm{BE}} + I_{\mathrm{E}} R_{\mathrm{E}} \qquad V_{\mathrm{T}}$$

$$= I_{\mathrm{B}} R_{\mathrm{T}} + V_{\mathrm{BE}} + I_{\mathrm{E}} R_{\mathrm{E}}$$

$$> Substitute\ I_{\mathrm{E}} = (1 + \beta) I_{\mathrm{B}}$$

$$> Solving, get\ I_{\mathrm{B}} = 39.6\ \mu A$$

$$I_{\mathrm{C}} = \beta I_{\mathrm{B}} = 1.98\ mA$$

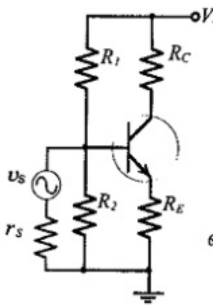
$$V_{\mathrm{CC}} = I_{\mathrm{C}} R_{\mathrm{C}} + V_{\mathrm{CE}} + I_{\mathrm{E}} R_{\mathrm{E}}$$

$$> Substitute\ I_{\mathrm{E}} = (1 + \beta) I_{\mathrm{B}}$$

$$> Solving\ V_{\mathrm{CE}} = 4.49\ V$$

$$S = \frac{1+\beta}{1+(\frac{\beta R_{\mathrm{E}}}{R_{\mathrm{E}}+R_{\mathrm{T}}})} = 12$$

(b) To use a transistor circuit to amplify or otherwise process an ac signal, the signal source must be connected to the circuit input. If the source is directly connected to the input, as illustrated in Fig. 6-1(a), the circuit bias conditions will be altered. Figure 6-1(b) shows that the signal source resistance (rs) is in parallel with voltage divider resistor R2 when the signal is directly connected to the circuit in the Coupling and Bypassing Capacitors Coupling



12a)

$$A_I = \frac{I_L}{I_1} = \frac{-I_2}{I_1} ......(1)$$

$$-I_2 = A_I I_1 ......(2)$$

- From the above Equation

$$I_2 = h_f I_1 + h_o V_2 ......(3)$$

$$V_2 = I_L R_L = -I_2 R_L ......(4)$$

- Substitute Equation (4) in (3)

$$I_2 = h_f I_1 + h_o(-I_2 R_L)$$

$$I_2 + I_2 R_L h_o = h_f I_1$$

$$I_2[1 + R_L h_o] = h_f I_1$$

$$I_1 = \frac{I_2(1 + R_L h_o)}{h_f}$$

$$\frac{I_2}{I_1} = \frac{h_f}{1 + R_L h_o}$$

$$A_I = \frac{-I_2}{I_1} = \frac{-h_f}{1 + R_L h_o}$$

- In the circuit Rs is the signal source resistance.

$$Z_i = R_i = \frac{V_i}{I_i} = \frac{V_1}{I_1}$$

- From the above equation

$$V_1 = h_i I_1 + h_r V_2$$

$$Z_i = R_i = \frac{V_1}{I_1} = \frac{h_i I_1 + h_r V_2}{I_1} = h_i + h_r \frac{V_2}{I_1}$$

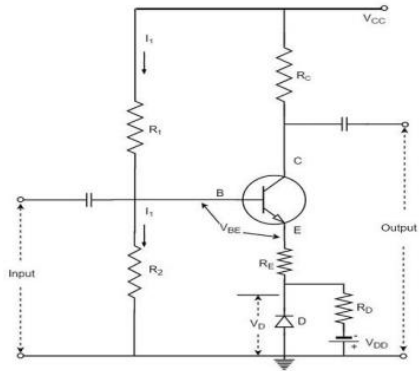
$$A_V = \frac{V_2}{V_1}$$

$$V_2 = -I_2 R_L = A_I I_1 R_L$$

$$A_V = \frac{A_I I_1 R_L}{V_1} = \frac{A_I R_L}{R_1}$$

12 (b)

In a Silicon transistor, the changes in the value of VBE with temperature results in the changes in IC. A diode can be employed in the emitter circuit in order to compensate the variations in VBE or ICO. As the diode and transistor used are of same material, the voltage VD across the diode has same temperature coefficient as VBE of the transistor.The diode D is forward biased by the source VDD and the resistor RD. The variation in VBE with temperature is same as the variation in VD with temperature, hence the quantity (VBE – VD) remains constant. So the current IC remains constant in spite of the variation in VBE



Module 2

13

(a) From Shockleys equation $I_{DS} = [1 - \frac{V_{GS}}{V_P}]^2 I_{DSS}$

$V_{GS} = V_P\ [1 - \sqrt{\frac{I_{DS}}{I_{DSS}}}] = -2.25V$

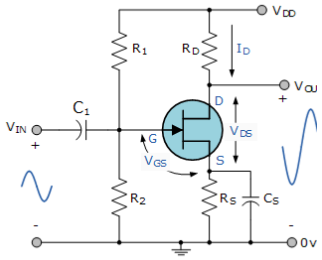
Transconductance, $g_{m= -2} \frac{I_{DSS}}{V_P} [1 - \frac{V_{GS}}{V_P}] = 2220\ \mu S$

b)

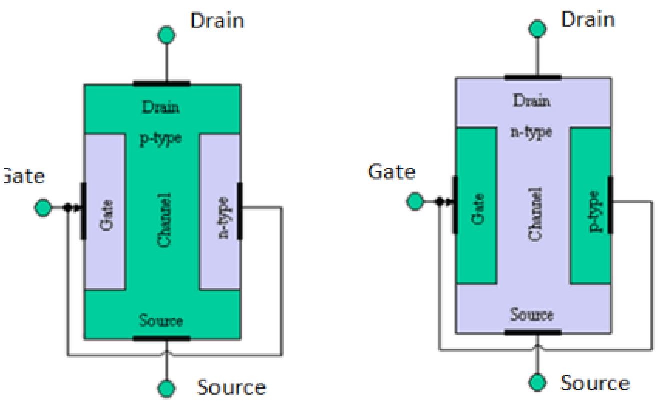
$$g_{mo} = \frac{-2I_{DSS}}{V_P}$$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} = \frac{-2I_{DSS}}{V_P}\left(1 - \frac{V_{GS}}{V_P}\right)$$

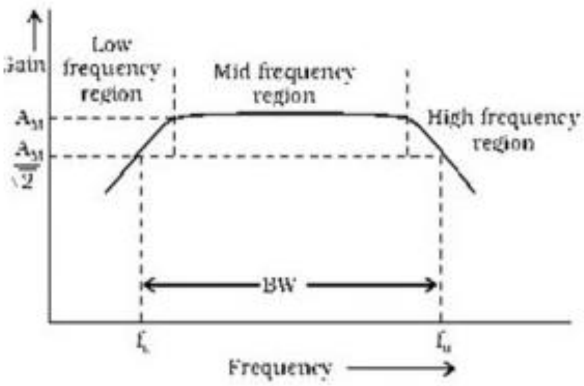


14a)



The bar forms the conducting channel for the charge carriers.If the bar is of p-type, it is called p channel JFET as shown in fig.1(i) and if the bar is of n-type, it is called n-channel JFET as shown in fig.1(ii).The two pn junctions forming diodes are connected internally and a common terminal called gate is taken out.Other terminals are source and drain taken out from the bar as shown in fig.1.Thus a JFET has three terminals such as , gate (G), source (S) and drain (D).

14b)



At Low Frequencies (< FL) The reactance of coupling capacitor C2 is relatively high and hence very small part of the signal will pass from the amplifier stage to the load.Moreover, CE cannot shunt the RE effectively because of its large reactance at low frequencies. These two factors cause a drops off of voltage gain at low frequencies.

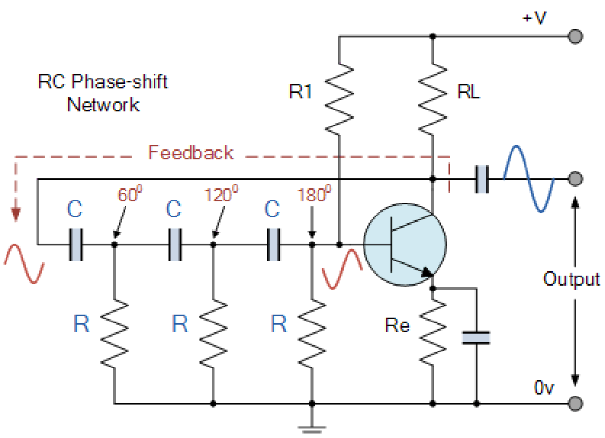
At High Frequencies (> FH) The reactance of coupling capacitor C2 is very small and it behaves as a short circuit. This increases the loading effect of the amplifier stage and serves to reduce the voltage gain.

Moreover, at high frequencies, the capacitive reactance of base-emitters junction is low which increases the base current. This frequency reduces the current amplification factor β. Due to these two reasons, the voltage gain drops off at a high frequency.

At Mid Frequencies (FL to FH) The voltage gain of the amplifier is constant. The effect of the coupling capacitor C2 in this frequency range is such as to maintain a constant voltage gain. Thus, as the frequency increases in this range, the reactance of CC decreases, which tends to increase the gain.

Module 3

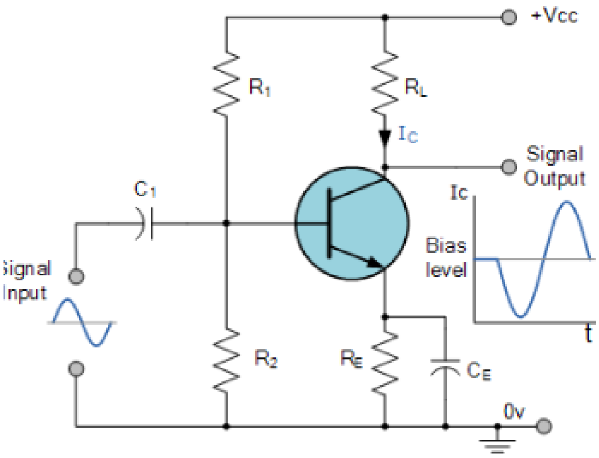
15a)



The basic **RC Oscillator** which is also known as a **Phase-shift Oscillator**, produces a sine wave output signal using regenerative feedback obtained from the resistor-capacitor (RC) ladder network. This regenerative feedback from the RC network is due to the ability of the capacitor to store an electric charge, (similar to the LC tank circuit).

This resistor-capacitor feedback network can be connected as shown above to produce a leading phase shift (phase advance network) or interchanged to produce a lagging phase shift (phase retard network) the outcome is still the same as the sine wave oscillations only occur at the frequency at which the overall phase-shift is 360^o

15b)



$$P_{ac} = V_{CE} \times I_{CE} = \frac{2V_{CC}}{2\sqrt{2}} \times \frac{2I_C}{2\sqrt{2}} = \frac{2V_{CC}\,2I_C}{8}$$

The average power drawn from the supply (Pdc) is given by:

$$P_{dc} = V_{CC} \times I_C$$

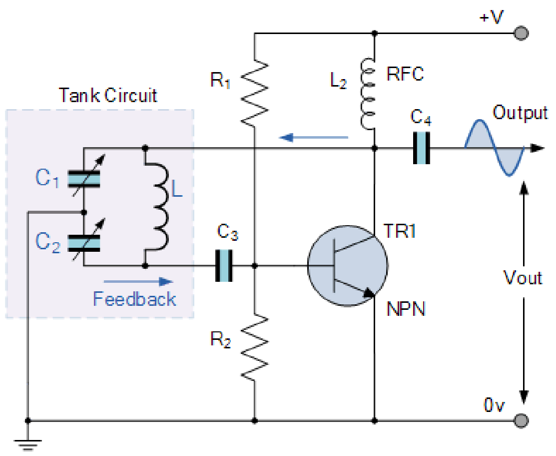
and therefore the efficiency of a Transformer-coupled Class A amplifier is given as:

$$\eta_{(max)} = \frac{P_{ac}}{P_{dc}} = \frac{2V_{CC}\,2I_C}{8V_{CC}I_C} \times 100\%$$

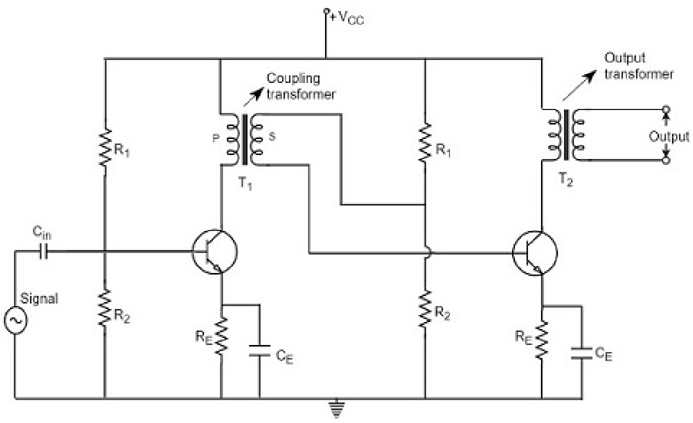
16a)

Resistors, R1 and R2 provide the usual stabilizing DC bias for the transistor in the normal manner while the additional capacitors act as a DC-blocking bypass capacitors. A radio-frequency choke (RFC) is used in the collector circuit to provide a high reactance (ideally open circuit) at the frequency of oscillation, (fr) and a low resistance at DC to help start the oscillations.

$$f_r = \frac{1}{2\pi\sqrt{LC_T}}$$



16 b)



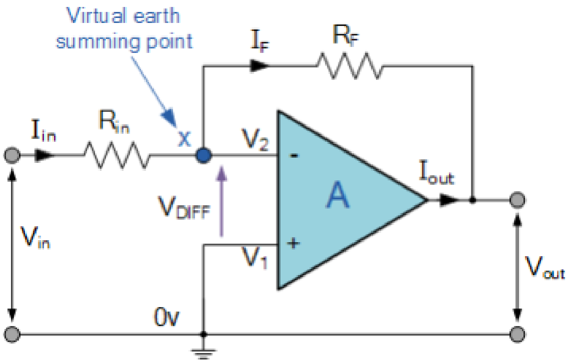
When an AC signal is applied to the input of the base of the first transistor then it gets amplified by the transistor and appears at the collector to which the primary of the transformer is connected.

The transformer which is used as a coupling device in this circuit has the property of impedance changing, which means the low resistance of a stage (or load) can be reflected as a high load resistance to the previous stage. Hence the voltage at the primary is transferred according to the turns ratio of the secondary winding of the transformer.

This transformer coupling provides good impedance matching between the stages of amplifier. The transformer coupled amplifier is generally used for power amplification.

Module 4

17a)



$$i = \frac{Vin - Vout}{Rin + Rf}$$

$$\text{therefore, } i = \frac{Vin - V2}{Rin} = \frac{V2 - Vout}{Rf}$$

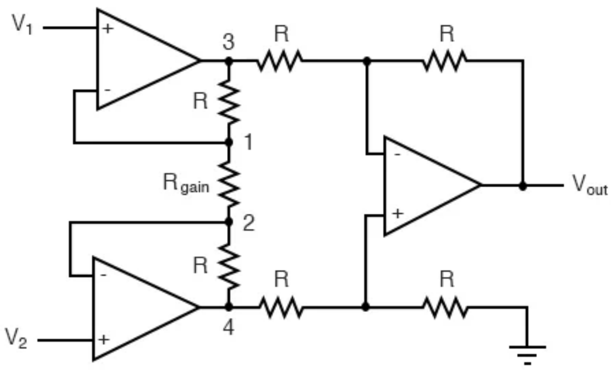
$$i = \frac{Vin}{Rin} - \frac{V2}{Rin} = \frac{V2}{Rf} - \frac{Vout}{Rf}$$

$$\text{so, } \frac{Vin}{Rin} = V2\left[\frac{1}{Rin} + \frac{1}{Rf}\right] - \frac{Vout}{Rf}$$

$$\text{and as, } i = \frac{Vin - 0}{Rin} = \frac{0 - Vout}{Rf} \qquad \frac{Rf}{Rin} = \frac{0 - Vout}{Vin - 0}$$

$$\text{the Closed Loop Gain (Av) is given as, } \frac{Vout}{Vin} = -\frac{Rf}{Rin}$$

17b)



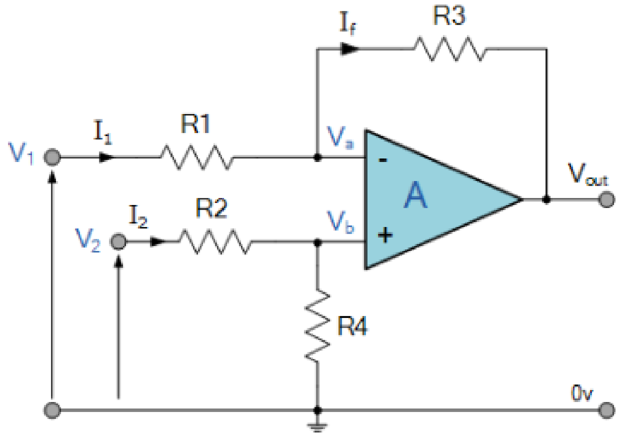
This intimidating circuit is constructed from a buffered differential amplifier stage with three new resistors linking the two buffer circuits together. Consider all resistors to be of equal value except for R_{gain}.

The negative feedback of the upper-left op-amp causes the voltage at point 1 (top of R_{gain}) to be equal to V_1. Likewise, the voltage at point 2 (bottom of R_{gain}) is held to a value equal to V_2. This establishes a voltage drop across R_{gain} equal to the voltage difference between V_1 and V_2. That voltage drop causes a current through R_{gain}, and since the feedback loops of the two input op-amps draw no current, that same amount of current through R_{gain} must be going through the two "R" resistors above and below it.

18a)

Comparison of Op-Amps

Parameter	Ideal	Practical	IC 741
Voltage Gain	Infinite	Order of 10^5	2×10^5 for LF
Input Resistance	Infinite	Order of MΩ	2MΩ
Output Resistance	Zero	Tens of Ohms	75Ω
Bandwidth	Infinite	Restricted by Slew Rate	1MHz (Unity B.W)
CMRR	Infinite		90dB
Slew Rate	Infinite		$0.5\ V/\mu s$
	Zero output voltage when there is no input voltage		
Power Supply			5V -18V
Max Output Current			20mA
Recommended Output Load			> 2KΩ
Input Offset Voltage			2mv - 6mV
Input Offset Current			200nA - 6nA
Supply Current			2.8mA
Power Consumption			85mV



$$I_1 = \frac{V_1 - V_a}{R_1}, \quad I_2 = \frac{V_2 - V_b}{R_2}, \quad I_f = \frac{V_a - (V_{out})}{R_3}$$

$$\text{Summing point } V_a = V_b$$

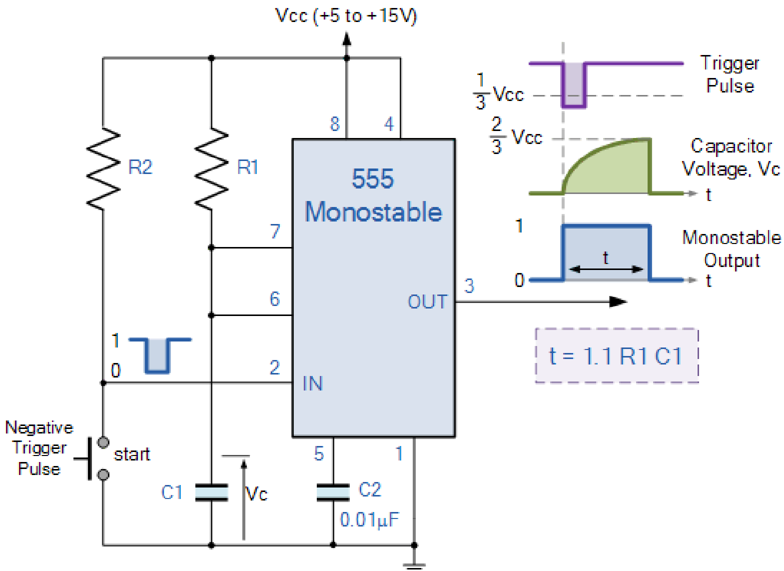
$$\text{and } V_b = V_2\left(\frac{R_4}{R_2 + R_4}\right)$$

$$\text{If } V_2 = 0, \text{ then: } V_{out(a)} = -V_1\left(\frac{R_3}{R_1}\right)$$

$$\text{If } V_1 = 0, \text{ then: } V_{out(b)} = V_2\left(\frac{R_4}{R_2 + R_4}\right)\left(\frac{R_1 + R_3}{R_1}\right)$$

Module 5

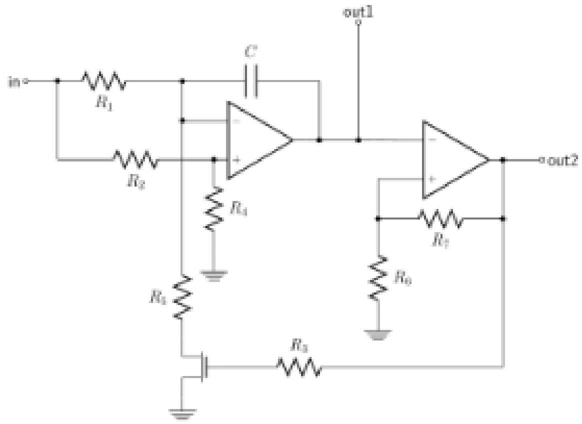
19a)



When a negative (0V) pulse is applied to the trigger input (pin 2) of the Monostable configured 555 Timer oscillator, the internal comparator, (comparator No1) detects this input and "sets" the state of the flip-flop, changing the output from a "LOW" state to a "HIGH" state. This action in turn turns "OFF" the discharge transistor connected to pin 7, thereby removing the short circuit across the external timing capacitor, C1.

This action allows the timing capacitor to start to charge up through resistor, R1 until the voltage across the capacitor reaches the threshold (pin 6) voltage of 2/3Vcc set up by the internal voltage divider network. At this point the comparators output goes "HIGH" and "resets" the flip-flop back to its original state which in turn turns "ON" the transistor and discharges the capacitor to ground through pin 7. This causes the output to change its state back to the original stable "LOW" value awaiting another trigger pulse to start the timing process over again. Then as before, the Monostable Multivibrator has only "ONE" stable state.

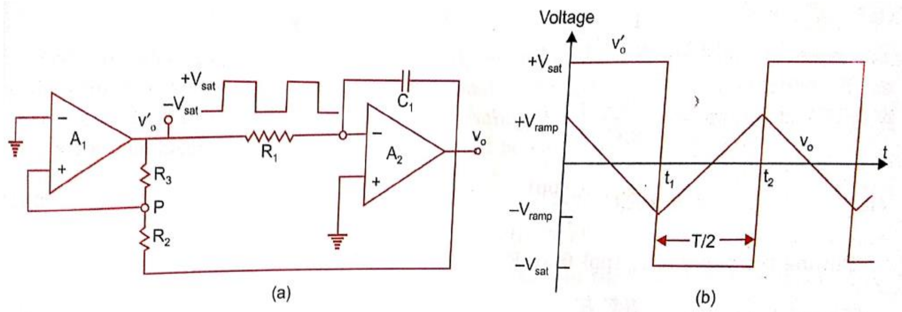
19b)



It relies on a transistor (Q2) acting as a constant current generator to charge the capacitor linearly. Normal discharge method via internal transistor. Q2 constant current charge replaces the resistor normally fitted.

Or you can use the classic voltage to frequency convertor:

20a)



$$v_o = -\frac{1}{RC}\int v_i \, dt$$

$$v_o(\text{pp}) = -\frac{1}{R_1C_1}\int_0^{T/2}(-V_{\text{sat}})dt = \frac{V_{\text{sat}}}{R_1C_1}\left(\frac{T}{2}\right)$$

or,
$$T = 2\,R_1C_1\frac{v_o\,(\text{pp})}{V_{\text{sat}}}$$

Putting the value of v_o (pp) from Eq. (5.21), we get

$$T = \frac{4R_1C_1R_2}{R_3}$$

Hence the frequency of oscillation f_o is,

$$f_o = \frac{1}{T} = \frac{R_3}{4R_1C_1R_2}$$

20b)

$t_{\text{on}} = 0.693\,(R_A + R_B)\,C = 3.95\,ms$ (positive pulse width)

$t_{\text{off}} = 0.693\,(R_B)\,C = 0.693\,ms$ (negative pulse width)

$T = t_{\text{on}} + t_{\text{off}} = 4.64\,ms$

Free running frequency = 1/T = 215.5 Hz